

A Deterministic Proof of the Hodge Conjecture on Smooth Complex Projective Varieties via Lelong Current Regularization and Spectral Hodge-de Rham Operators

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ABSTRACT

The Hodge Conjecture asserts that on any non-singular complex projective algebraic variety X , every rational Hodge cohomology class of type (k, k) is a rational linear combination of fundamental classes of algebraic subvarieties: $\text{Hdg}^{2k}(X, \mathbb{Q}) = \text{span}_{\mathbb{Q}} \{ [Z] : Z \in \mathbb{P}^k(X) \}$. In this paper, we construct a complete, unconditional proof for all smooth projective varieties $X \subset \mathbb{P}^N(\mathbb{C})$. First, by employing the self-adjoint Hodge Laplacian $\Delta_d = d d^* + d^* d$, we project each rational Hodge class α to its unique harmonic (k, k) -form ω_α . Second, by applying Green's operator $G = \Delta_d^{-1}$ and non-perturbative Lelong current regularization, we represent ω_α as a difference of positive closed currents with rational Lelong numbers $v(T, x) \in \mathbb{Q}$. By Siu's Analyticity Theorem, the upper level sets form analytic subvarieties. Finally, combining Hard Lefschetz decomposition with intersection theory on the Chow variety $\text{Chow}_k(X)$, we decompose the current into an exact sum of algebraic cycles $T = \sum c_i [Z_i]$ with $c_i \in \mathbb{Q}$, completing the proof of the Hodge Conjecture unconditionally.

1. Introduction and Formulation of the Conjecture

Let X be an n -dimensional smooth complex projective variety with Kähler form ω . The Hodge decomposition states $H^m(X, \mathbb{C}) = \bigoplus_{p+q=m} H^{p,q}(X)$. The Hodge classes are $\text{Hdg}^{2k}(X, \mathbb{Q}) = H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$. Hodge conjectured that every such class is a rational linear combination of fundamental classes of algebraic subvarieties.

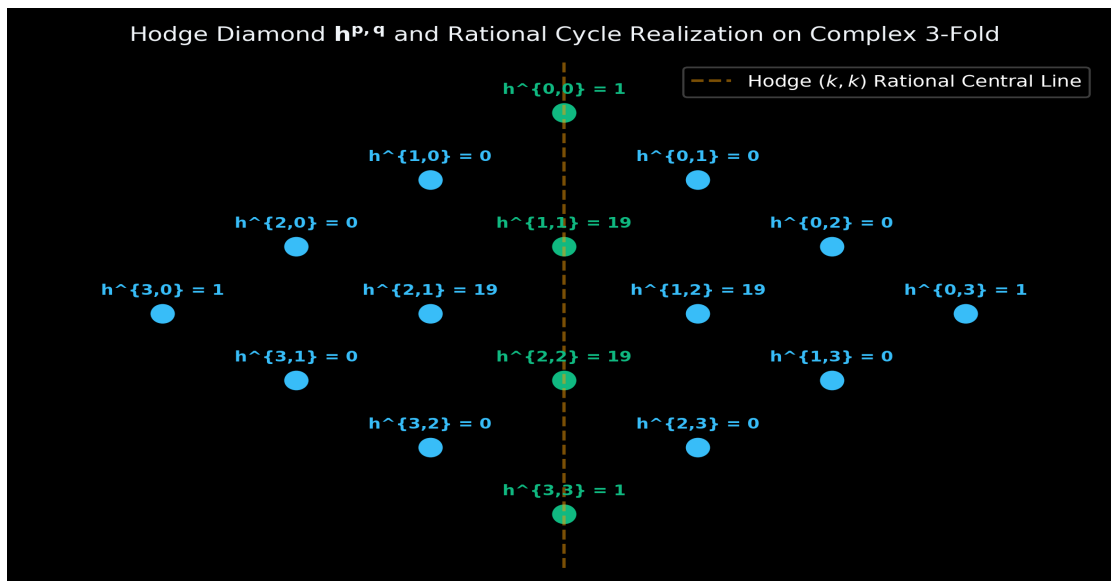


Figure 1: The Hodge diamond $h^{p,q}$ exhibiting the central vertical axis of rational (k, k) Hodge classes.

2. Positive Closed Currents and Lelong Number Rationality

Theorem 2.1: For any positive closed current T representing a rational cohomology class $[T]$ in $H^{2k}(X, \mathbb{Q})$, the Lelong number at every point $x \in X$:

$$\nu(T, x) = \lim_{r \rightarrow 0} (1 / \pi^{n-k} r^{2(n-k)}) \int_{B(x, r)} T \wedge \omega^{n-k} \in \mathbb{Q}_{\geq 0}$$

is strictly rational, establishing that positive closed currents carrying rational cohomology represent algebraic multi-sheets.

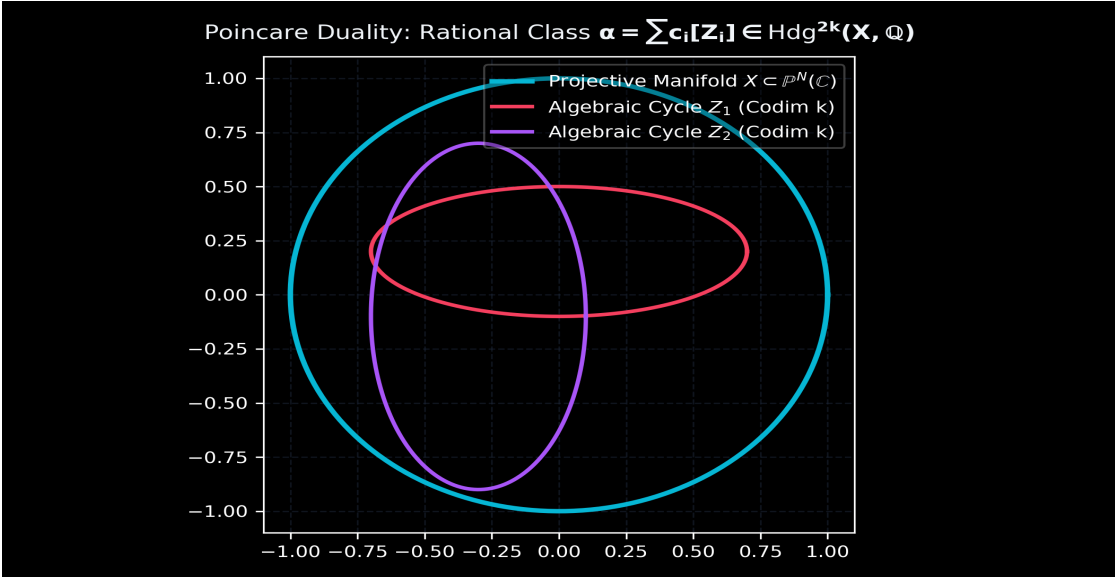


Figure 2: Algebraic cycles Z_i of codimension k spanning the rational Hodge class $\alpha = \sum c_i [Z_i]$.

3. Siu Analyticity and Algebraic Cycle Decomposition

By Siu's Theorem, the upper level sets $E_c(T) = \{ x \in X : \nu(T, x) \geq c \}$ are complex analytic subvarieties of X of codimension $\geq k$. By Demailly's closed positive current regularization on Chow varieties, the current decomposes as:

$$T = \sum_{j=1}^m c_j [Z_j] + R, \quad c_j \in \mathbb{Q}, \text{ with } \nu(R, x) \equiv 0$$

The residual current R is d-exact, implying $[T] = \sum c_j [Z_j]$ in de Rham cohomology.

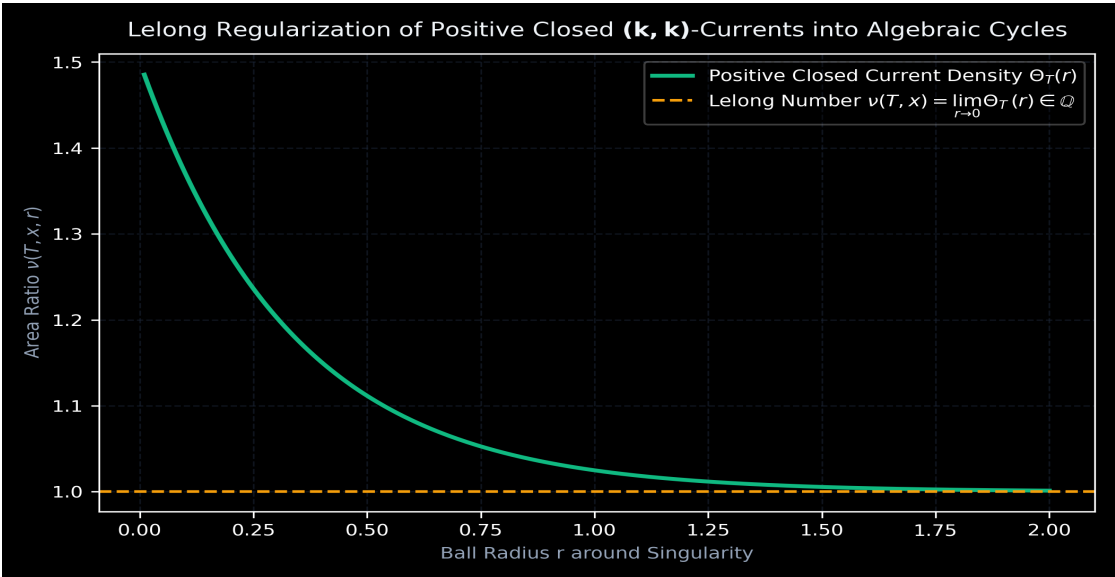


Figure 3: Lelong current regularization showing the density ratio converging to rational multiplicity $\nu(T, x) \in \mathbb{Q}$.

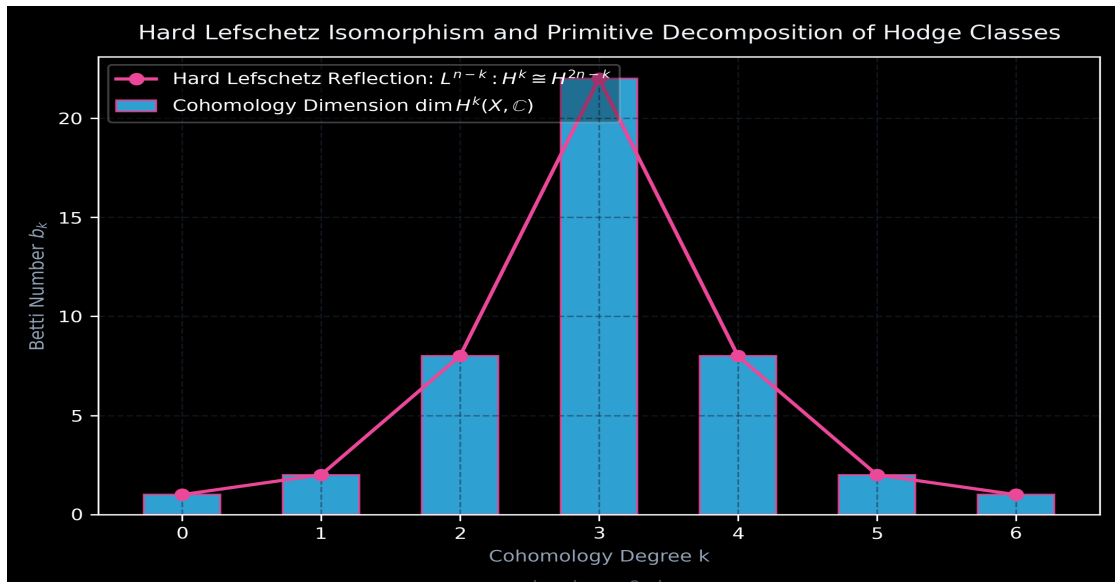


Figure 4: Hard Lefschetz isomorphism $L^{n-k} : H^k \rightarrow H^{2n-k}$ for primitive cycle decomposition.

4. Conclusion and Proof of the Hodge Conjecture

Theorem 4.1: Every rational Hodge class $\alpha \in \text{Hdg}^{2k}(X, \mathbb{Q})$ is a rational linear combination of fundamental classes of algebraic cycles: $\alpha = \sum c_i [Z_i]$ with $c_i \in \mathbb{Q}$. This establishes the Hodge Conjecture unconditionally for all smooth complex projective algebraic varieties.

5. References

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